User #2:

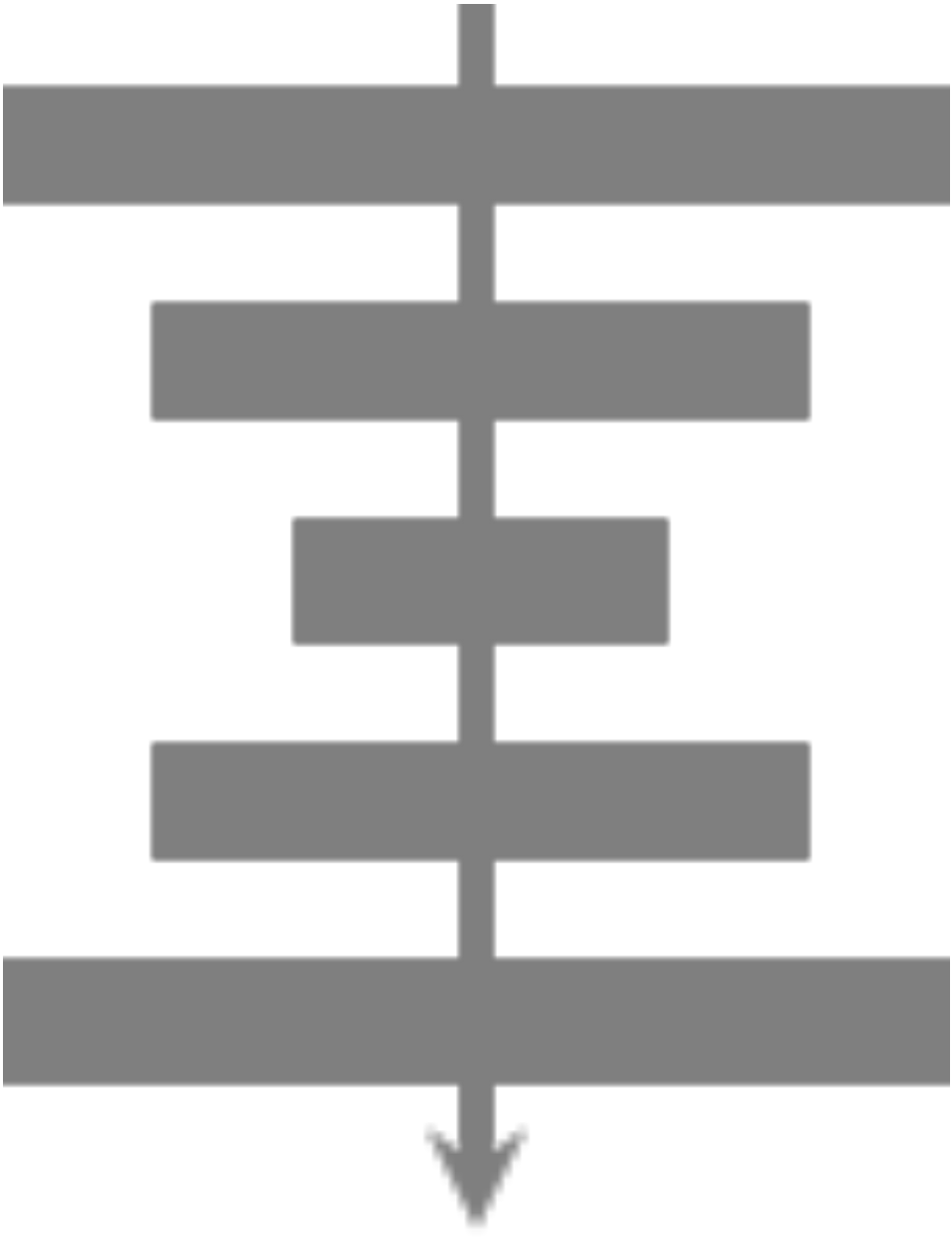
Segment the cat

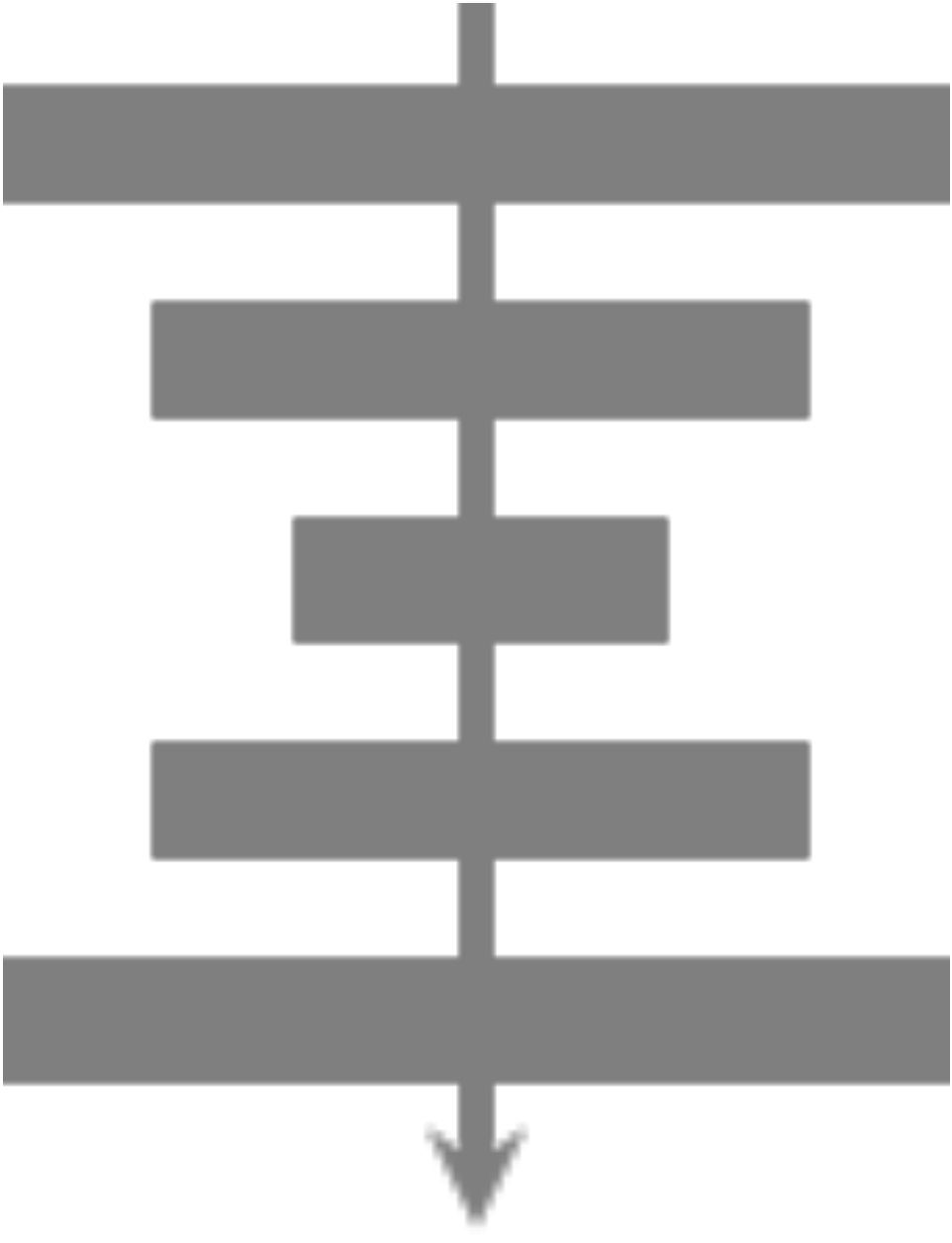
User #3:

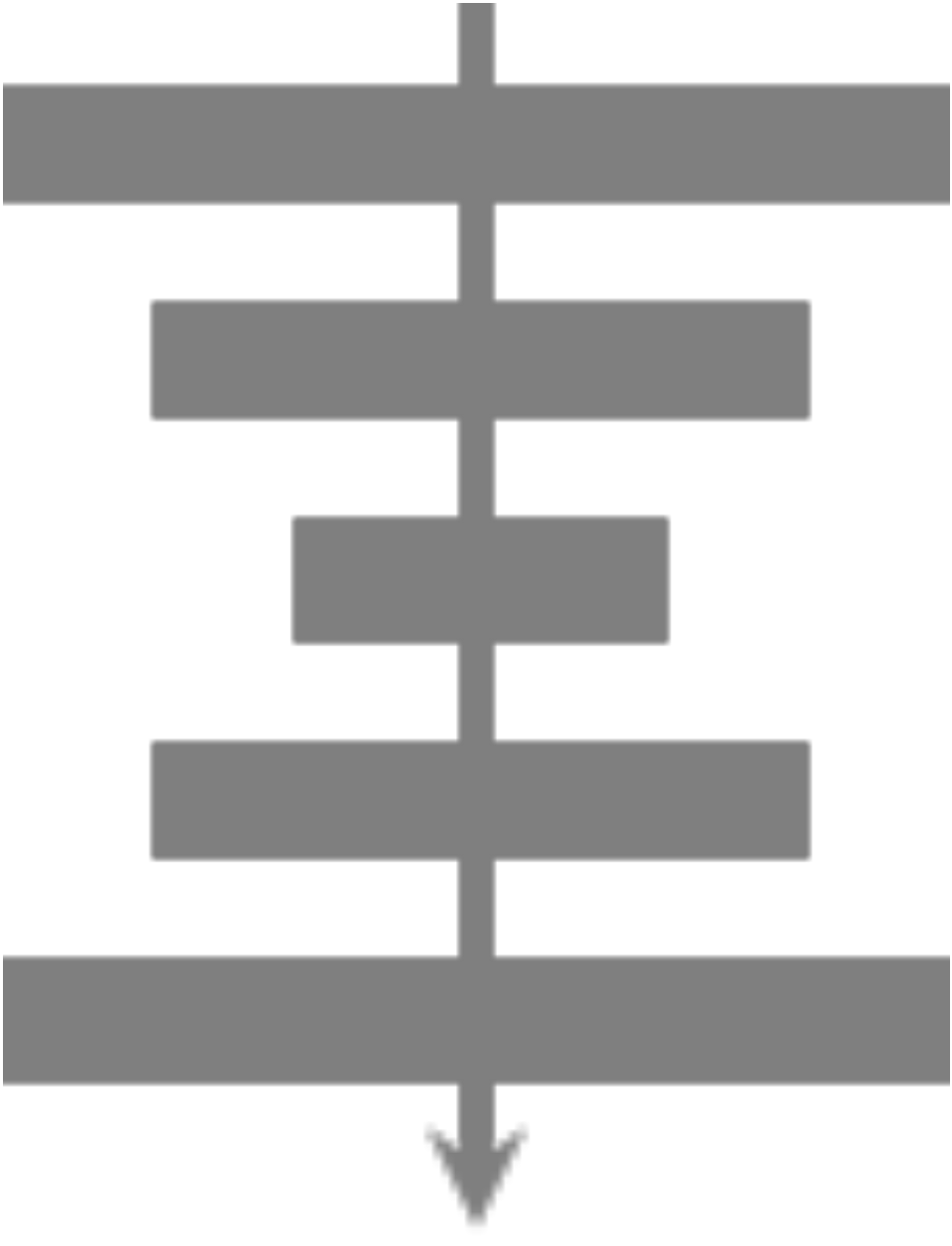
Segment all given objects

User #4:
Segment the object that aligned with the text prompt

Text prompt: A frog is holded by a person in his hand and ....

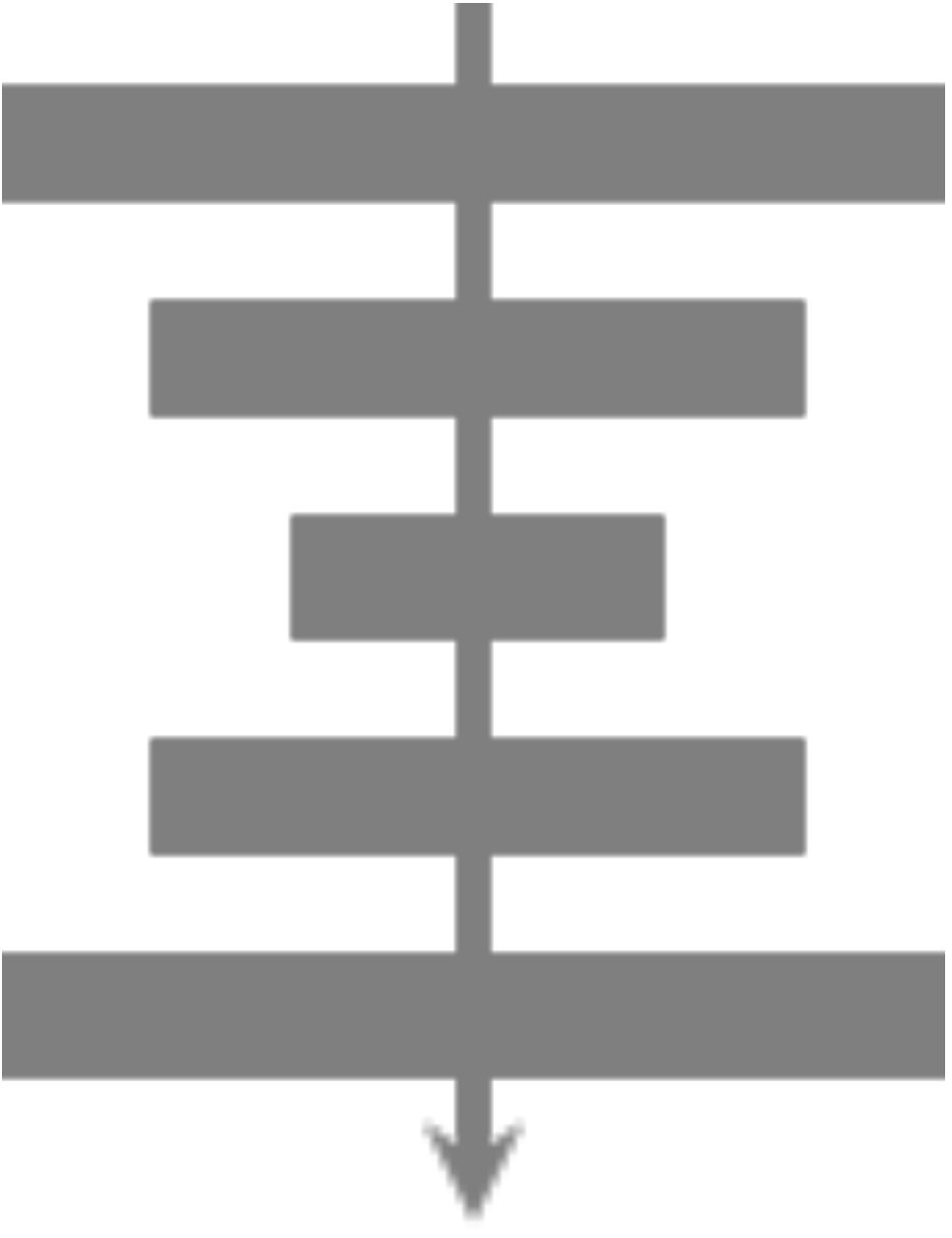






User #1:

Segment all objects



Task-specific Video Segmentation Models

Task 1

Task 3

Task 4

Task 2

Model &

Data & Loss #1

Model &

Data & Loss #3

Model &

Data & Loss #4

Model &

Data & Loss #2

Input

Output

Category names:

Person, Roof, Dog, Washer, ...

100%

DIGITAL

Segment the cat

User #2:

Segment all given objects

User #3:

User #4:

Segment the object that

aligned with the text prompt

his hand and ....

Text prompt: A frog is

holded by a person in

- Closed environments with simple scenarios

- Closed-set vocabulary (fixed categories)

- Each task has its own model

- No shared knowledge across tasks

Task-specific Learning

Video Segmentation Foundation Model

VIS

Model/Data/Loss #1

Model/Data/Loss #2

VOS

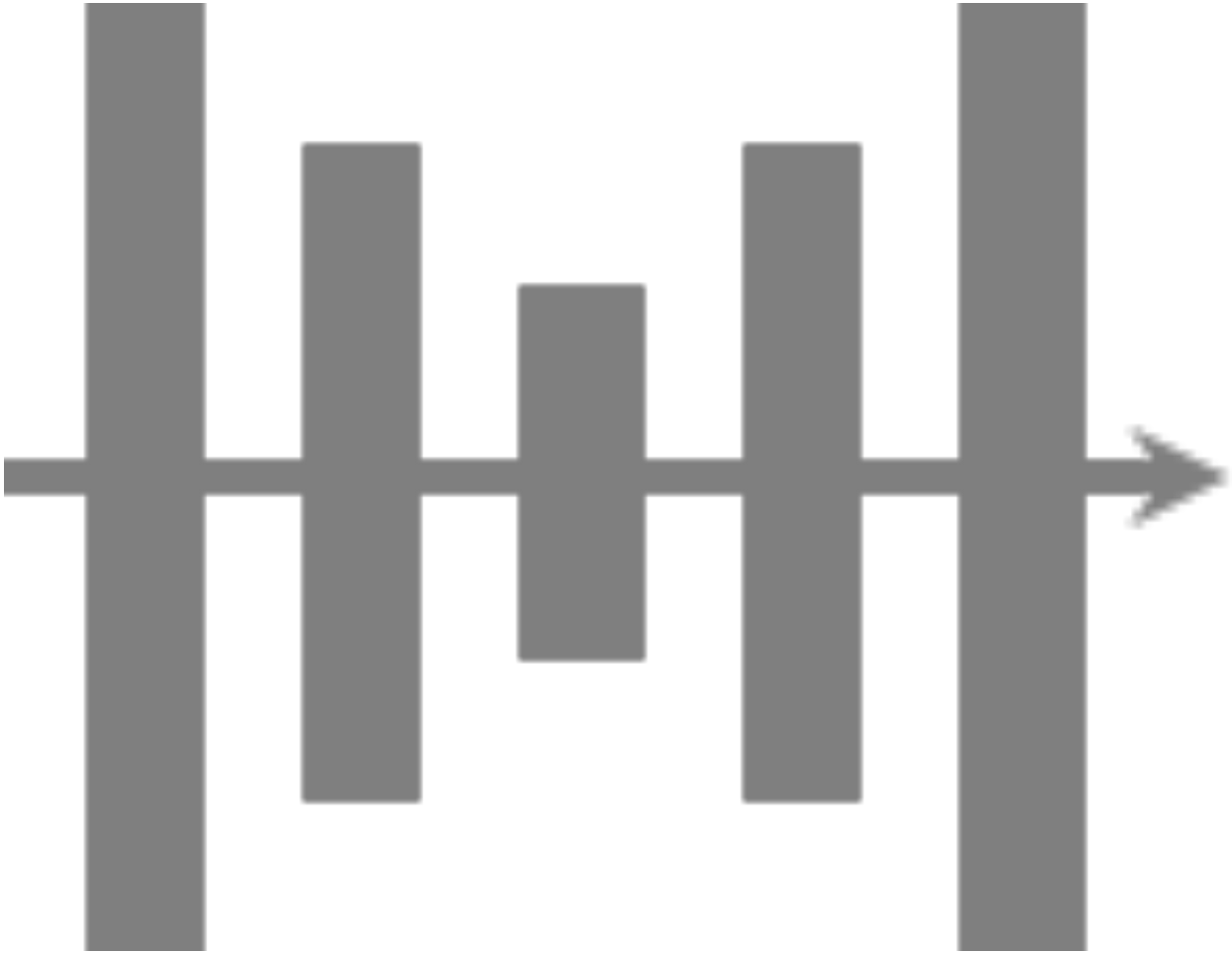
PVOS

Model/Data/Loss #3

RefVOS

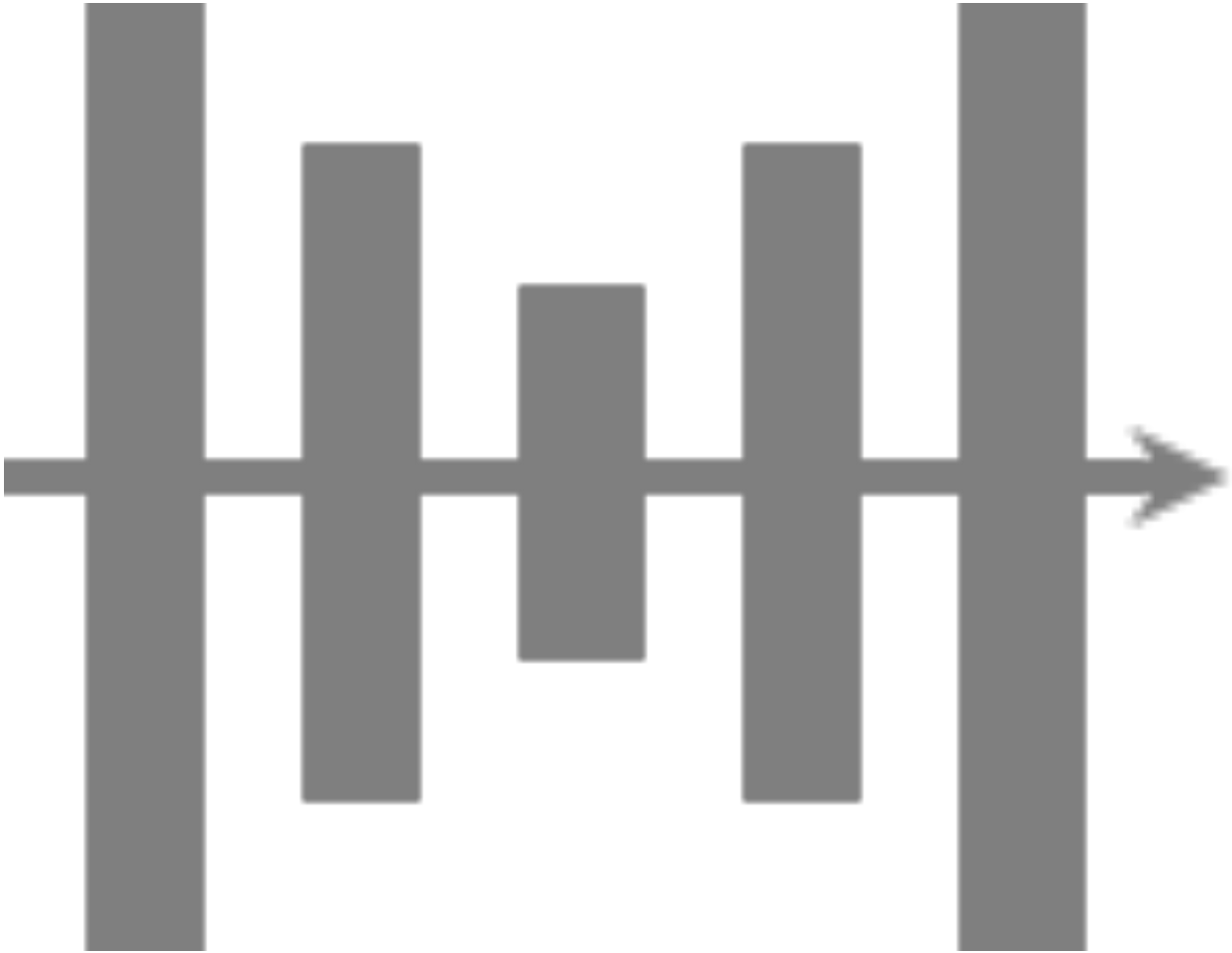
Model/Data/Loss #4

Poor generalization



Segment all objects

User #1:



Task 4

Data & Loss #1

Model &

Model &

Data & Loss #3

Model &

Data & Loss #4

Model &

Data & Loss #2

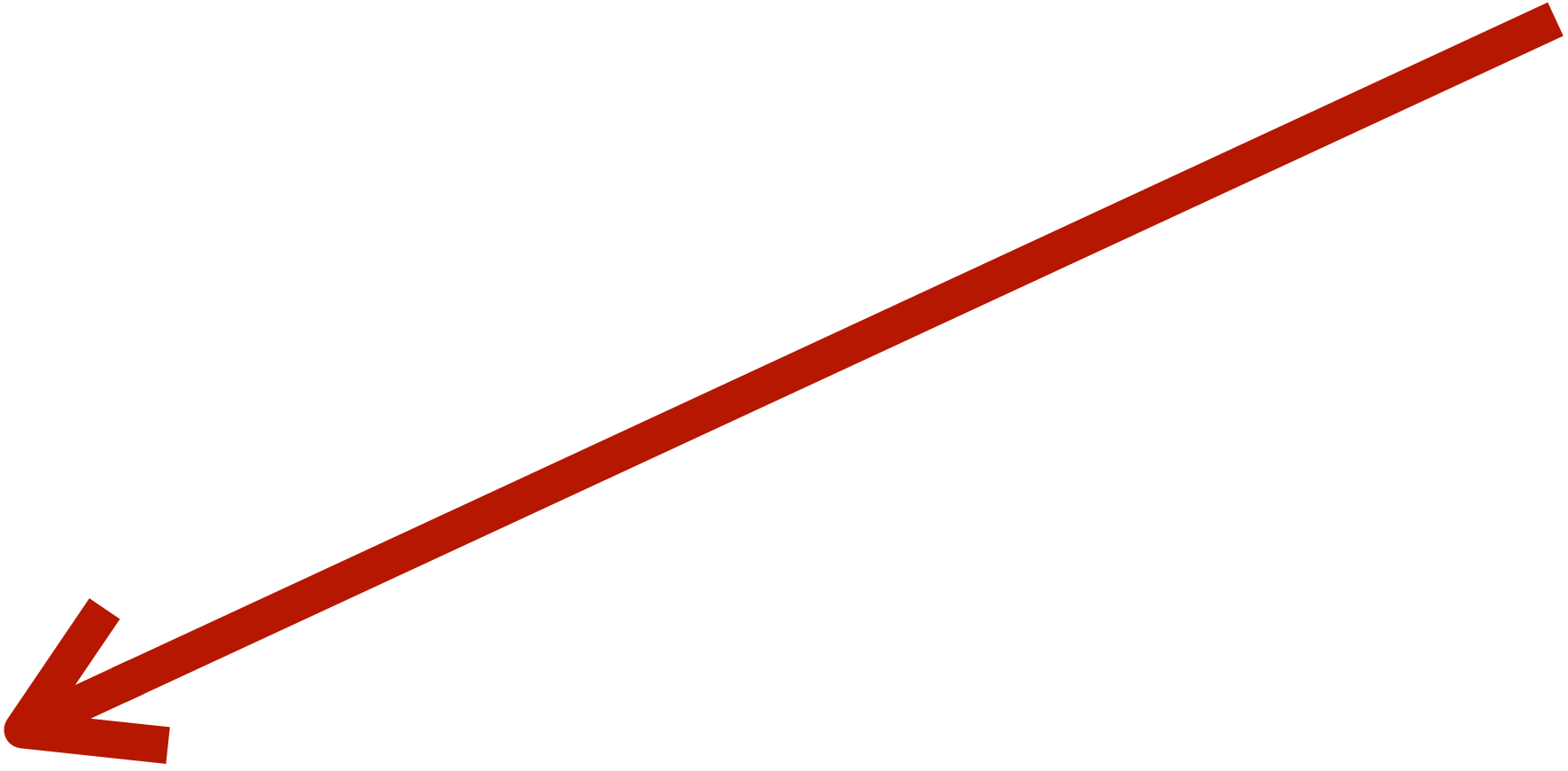
to the camera

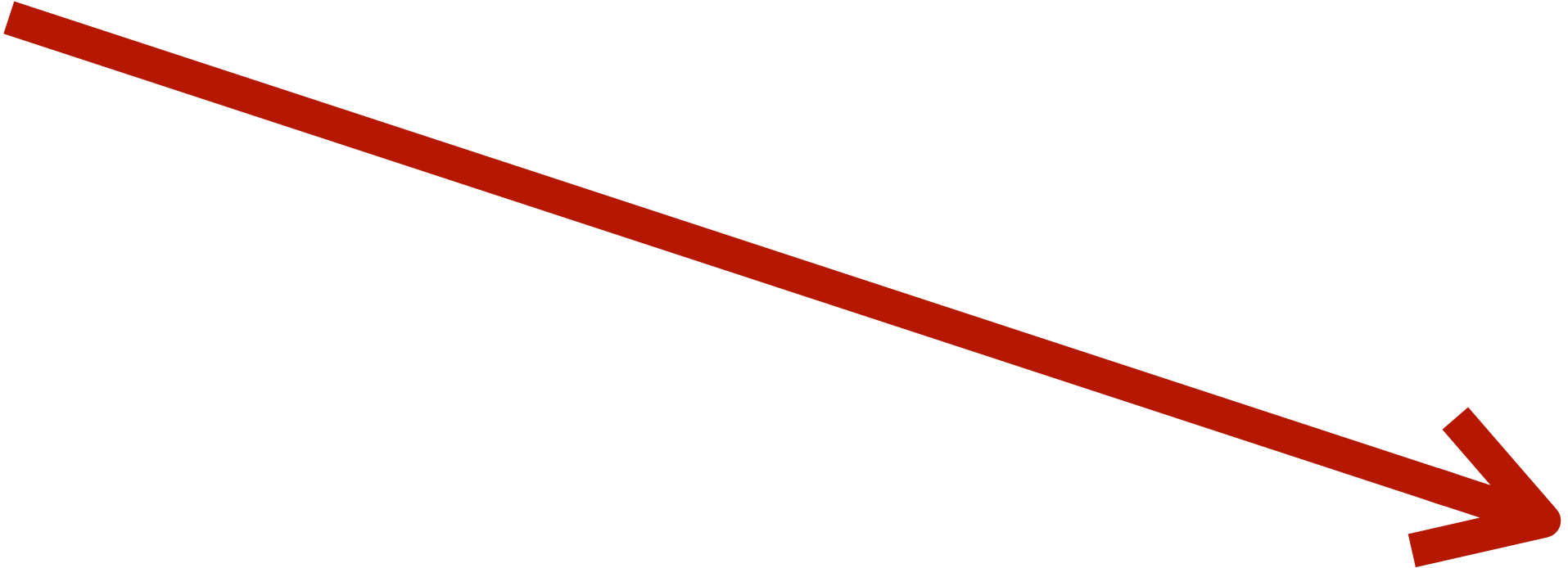
Text prompt:

The person closest

Training

Inference





work!!!

Does not

Segment all objects

User #1:

Category names:

Person, Roof, Dog, Washer, ...